Unit 1 Look Back · Biochemistry

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Rate yourself on each target

Target	Got it	Shaky	Lost
LT 1.1 Explain why water is polar and how hydrogen bonds give water its properties.			
LT 1.2 Describe the four macromolecules: elements, monomers, examples, jobs; how they are built and broken.			
LT 1.3 Label a reaction and its energy graph and explain how enzymes and conditions change the rate.			

Seven questions

1. (LT 1.1 · SOL Challenge Q2) The raindrop on the window: which bonds hold each molecule together, and which hold the drop together?

2. (LT 1.2 · SOL Challenge Q9) A building block paired with the chain it makes.

3. (LT 1.3 · SOL Challenge Q14) Hydrogen peroxide on a shelf versus with a slice of potato.

4. (LT 1.1 · Unit 1 Practice OB-U01-02) In a glass of water, where does a hydrogen bond form?
   (A) Between the oxygen and a hydrogen inside the same water molecule
   (B) Between the slightly positive hydrogen of one water molecule and the slightly negative oxygen of a neighboring molecule
   (C) Between the oxygen atoms of two neighboring water molecules
   (D) Between the hydrogen atoms of two neighboring water molecules

5. (LT 1.1 · Unit 1 Practice OB-U01-07) Why does solid water (ice) float on liquid water, when most solids sink in their own liquid?
   (A) Ice contains trapped air bubbles that lift it
   (B) Hydrogen bonds lock frozen water molecules into an open pattern, so ice is less dense than liquid water
   (C) Ice has more mass than the same volume of liquid water
   (D) Cold liquid water is less dense than ice

6. (LT 1.2 · Unit 1 Practice OB-U01-18) Which description fits a polymer?
   (A) A long molecule built by linking many smaller repeating units
   (B) A single small unit that can join with others
   (C) Any molecule that contains carbon
   (D) Any molecule that dissolves in water

7. (LT 1.3 · Unit 1 Practice OB-U01-35) On a graph of energy during a reaction, the line starts at the reactants, rises to a peak, and falls to the products. What is the height from the reactants' starting level up to the peak?
   (A) The energy of the products
   (B) The activation energy
   (C) The total energy released
   (D) The energy of the enzyme

Draw it from memory

Two water molecules joined by a hydrogen bond, with δ+, δ−, a polar covalent bond and the hydrogen bond labeled.

Fix the claim

“Enzymes get used up in the reactions they speed up, so a cell has to keep making new ones for every reaction.” Rewrite it so it is correct, and give one piece of evidence from the potato lab.

Fix one

Pick the question above you were least sure of. What was the idea you needed?

My plan

One thing I will do before the test (a Lesson Quiz, a deck, a Bio Tool, a question for Mr. Schwebach):

Check 1–3 (SOL Challenge Q2, Q9, Q14)

Check 4–7 (Unit 1 Practice)

Original practice questions written to the Virginia 2018 Biology SOL. Not produced or endorsed by the Virginia Department of Education.